Values

See "Women's values" at the beginning of section 3.D: Background (p. 74).

Resources

Costs of antacids vary widely, but generic products can be relatively low cost. Acupuncture requires professional training and skills and is likely to be associated with higher costs.

Equity

The prevalence of health-seeking behaviour and treatment for heartburn in pregnancy may be unequal among advantaged and disadvantaged women. However, it is not known whether interventions to relieve heartburn might impact inequalities.

Acceptability

Qualitative evidence from a range of LMICs suggests that women may be more likely to turn to traditional healers, herbal remedies or TBAs to treat these symptoms (moderate confidence in the evidence) *(22)*. In addition, evidence from a diverse range of settings indicates that while women generally appreciate the interventions and information provided during antenatal visits, they are less likely to engage with services if their beliefs, traditions and socioeconomic circumstances are ignored or overlooked by health-care providers and/or policy-makers (high confidence in the evidence). This may be particularly pertinent for an intervention like acupuncture, which may be culturally alien and/or poorly understood in certain contexts. Indirect evidence also indicates that women welcome the pregnancy-related advice and guidance given by health-care professionals during antenatal visits, so may respond to lifestyle suggestions favourably (moderate confidence in the evidence).

Feasibility

Qualitative evidence suggests that a lack of resources may limit the offer of treatment for this condition (high confidence in the evidence) *(45)*.

D.3: Interventions for leg cramps

RECOMMENDATION D.3: Magnesium, calcium or non-pharmacological treatment options can be used for the relief of leg cramps in pregnancy, based on a woman's preferences and available options. *(Recommended)*

Remarks

- The review found no evidence on the effect of non-pharmacological therapies, such as muscle stretching, relaxation, heat therapy, dorsiflexion of the foot and massage.
- The evidence on magnesium and calcium is generally of low certainty. However, the GDG agreed that they are unlikely to be harmful in the dose schedules evaluated in included studies.
- Further research into the etiology and prevalence of leg cramps in pregnancy, and the role (if any) of magnesium and calcium in symptom relief, is needed.

Summary of evidence and considerations

Effects of interventions for leg cramps compared with other, no or placebo interventions (EB Table D.3)

The evidence on the effects of various interventions for leg cramps in pregnancy is derived from a Cochrane review that included six small trials involving 390 pregnant women with leg cramps *(162)*. Three studies from Norway (42 women), Sweden (69 women) and Thailand (86 women) contributed data on oral magnesium compared with placebo. One study from Sweden (43 women) compared oral calcium with no treatment; a study conducted in the Islamic Republic of Iran (42 women) compared oral vitamins B6 and B1 with no treatment; and another conducted in Sweden compared oral calcium with vitamin C (30 women). Symptom relief, measured in different ways, was the primary outcome in these studies, and other maternal and perinatal outcomes relevant to this guideline were not reported.

Oral magnesium versus placebo

In three small studies, women in the intervention group were given 300–360 mg magnesium per day in two or three divided doses. Studies measured persistence or occurrence of leg cramps in different ways, so results could not be pooled. Moderate-certainty evidence from the Thai study suggests that women receiving magnesium are more likely to experience a 50% reduction in the number of leg cramps (1 trial, 86 women; RR: 1.42, 95% CI: 1.09–1.86). The same direction of effect was found in the Swedish study, which reported the outcome "no leg cramps" after treatment, but the evidence was of low certainty (1 trial, 69 women; RR: 5.66, 95% CI: 1.35–23.68). Low-certainty evidence suggests that oral magnesium has little or no effect on the occurrence of potential side-effects, including nausea, diarrhoea, flatulence and bloating. Evidence from the third study was judged to be very uncertain.

Oral calcium versus no treatment

Calcium, 1 g twice daily for two weeks, was compared with no treatment in one small study. Low-certainty evidence suggests that women receiving calcium treatment are more likely to experience no leg cramps after treatment (43 women; RR: 8.59, 95% CI: 1.19–62.07).

Oral calcium versus vitamin C

Low-certainty evidence suggests that there may be little or no difference between calcium and vitamin C in the effect (if any) on complete symptom relief from leg cramps (RR: 1.33, 95% CI: 0.53–3.38).

Oral vitamin B1 and B6 versus no treatment

One study evaluated this comparison, with 21 women receiving vitamin B1 (100 mg) plus B6 (40 mg) once daily for two weeks and 21 women receiving no treatment; however, the low-certainty findings are contradictory and difficult to interpret.

Additional considerations

- The review found no evidence on non-pharmacological therapies, such as muscle stretching, massage, relaxation, heat therapy and dorsiflexion of the foot.

Values

See "Women's values" at the beginning of section 3.D: Background (p. 74).

Resources

Magnesium and calcium supplements are relatively low-cost interventions, particularly when administered for limited periods of two to four weeks.

Equity

The potential etiology of leg cramps being related to a nutritional deficiency (magnesium) suggests that the prevalence of leg cramps might be higher in disadvantaged populations. In theory, therefore, nutritional interventions may have equity implications, but evidence is needed.

Acceptability

Qualitative evidence from a diverse range of settings suggests that women generally appreciate the pregnancy-related advice given by health-care professionals during ANC, so may respond to supplement suggestions favourably (moderate confidence in the evidence) *(22)*. Evidence from some LMICs suggests that women hold the belief that pregnancy is a healthy condition and may turn to traditional healers and/or herbal remedies to treat these kinds of associated symptoms (high confidence in the evidence).

Feasibility

Qualitative evidence suggests that a lack of resources may limit the offer of treatment for this condition (high confidence in the evidence) *(45)*. In addition, where there are additional costs for pregnant women associated with treatment, women are less likely to use it.

D.4: Interventions for low back and pelvic pain

RECOMMENDATION D.4: Regular exercise throughout pregnancy is recommended to prevent low back and pelvic pain. There are a number of different treatment options that can be used, such as physiotherapy, support belts and acupuncture, based on a woman's preferences and available options. *(Recommended)*

Remarks

- Exercise to prevent low back and pelvic pain in pregnancy can take place on land or in water. While exercise may also be helpful to relieve low back pain, it could exacerbate pelvic pain associated with symphysis pubis dysfunction and is not recommended for this condition.
- Regular exercise is a key component of lifestyle interventions, which are recommended for pregnant women as part of ANC to prevent excessive weight gain in pregnancy (see Recommendation A.9).
- Pregnant women with low back and/or pelvic pain should be informed that symptoms usually improve in the months after birth.
- Women should be informed that it is unclear whether there are side-effects to alternative treatment options due to a paucity of data.
- Standardized reporting of outcomes is needed for future research on treatment for low back and/or pelvic pain in pregnancy.

Summary of evidence and considerations

Effects of interventions for low back and pelvic pain compared with other, no or placebo interventions (EB Table D.4)

The evidence on the effects of various interventions for low back and pelvic pain in pregnancy was derived from a Cochrane review that included 34 trials involving 5121 women *(165)*. The definitions and terminology of low back and pelvic pain varied such that in 15 trials the interventions were aimed at reducing low back pain, in six trials interventions were for pelvic pain, and in 13 trials the interventions were for low back and pelvic pain. Most trials evaluated treatment; however, six trials evaluated prevention. Few trials contributed data to analyses and several individual study findings were described only in narrative. Main outcomes were relief of symptoms and functional disability, and perinatal outcomes relevant to this guideline were not reported. Comparisons included:

1. any exercise (plus standard care) versus standard care
2. acupuncture (plus standard care) versus sham acupuncture (plus standard care)
3. acupuncture (plus standard care) versus individualized physiotherapy (plus standard care)
4. osteopathic manipulation (plus standard care) versus standard care
5. one type of support belt versus another typee
6. multimodal interventions versus standard care.

Any exercise (plus standard care) versus standard care

Seven trials (645 women) contributed data to this comparison for low back pain. Trials were conducted in Brazil, the Islamic Republic of Iran, Norway, South Africa and Thailand. Exercise interventions varied from individually supervised exercise to group exercise, including yoga and aqua-aerobics, and some included education via CDs and booklets. Interventions ran for 8–12 weeks and the presence or intensity of pain was assessed in most trials using visual analogue scales. However, the evidence on symptom relief from a meta-analysis of these seven studies is very uncertain. Low-certainty evidence suggests that functional disability scores are better with exercise interventions for low back pain (2 trials, 146 women; standardized MD: 0.56 lower, 95% CI: 0.23–0.89 lower). Evidence on pain intensity (symptom scores) for low back pain was assessed as very uncertain.

Low-certainty evidence suggests that an 8- to 12-week exercise programme may reduce low back and pelvic pain compared with standard care (4 trials, 1176 women; RR: 0.66, 95% CI: 0.45–0.97) and moderate-certainty evidence shows that healthy pregnant women taking part in an exercise programme are probably less likely to take sick leave related to low back and pelvic pain (2 trials, 1062 women; RR: 0.76; 95% CI: 0.62–0.94).

Acupuncture (plus standard care) versus sham acupuncture (plus standard care)

Four small studies conducted in Sweden and the USA evaluated the effects of acupuncture plus standard care versus sham acupuncture plus standard care. However, little data were extracted from these studies and data could not be pooled. Low-certainty evidence from one study suggests that acupuncture may relieve low back and pelvic pain (72 women; RR: 4.16, 95% CI: 1.77–9.78). Evidence from other studies was variously reported and very uncertain.

Acupuncture (plus standard care) versus individualized physiotherapy (plus standard care)

One small study conducted in Sweden involving 46 women with low back and pelvic pain evaluated this comparison. Women's satisfaction with treatment was the main outcome, but the evidence was assessed as very uncertain.

Osteopathic manipulation therapy (OMT) (plus standard care) versus no osteopathic manipulation (standard care)

Three studies evaluated OMT; however, data could not be pooled and the evidence from individual studies is inconsistent. The largest study involving 400 women compared OMT plus standard care with placebo ultrasound plus standard care, or standard care only. Limited data from this study suggests that OMT may relieve low back pain symptoms more than standard care, and may lead to lower functional disability scores, but may not be better than placebo ultrasound for these outcomes.

One type of support belt versus another type

One small study conducted in Australia compared two types of support belts in womewith low back pain, the BellyBra® and Tubigrip® (N = 94) and the evidence from this study was assessed as very low-certainty evidence.

Multimodal interventions versus standard care

One study in the USA reported the effect of a multimodal intervention that included weekly manual therapy by a chiropractic specialist, combined with daily exercise at home, and education versus standard care (rest, exercise, heat pads and analgesics) on low back and pelvic pain. Moderate-certainty evidence suggests that the multimodal intervention is probably associated with better pain scores (1 study, 169 women; MD: 2.70 lower, 95% CI: 1.86–3.54 lower) and better functional disability scores (MD: 1.40 lower; 95% CI: 0.71–2.09 lower) compared with standard care.

Additional considerations

- It is not clear whether the evidence on exercise interventions applies equally to low back pain and pelvic pain, or equally to prevention and treatment, as data from studies of prevention and treatment were pooled. Evidence from two studies on the effect of exercise plus education suggests that such interventions may have little or no effect on preventing pelvic pain (RR: 0.97; 95% CI: 0.77–1.23).
- Very low-certainty evidence on a number of other interventions, such as transcutaneous electrical nerve stimulation (TENS), progressive muscle relaxation with music, craniosacral therapy, and acetaminophen (paracetamol) – which were evaluated in single small trials with apparent relief of symptoms relative to standard care – was also presented in the review.
- Standard care of low back and pelvic pain symptoms usually comprises rest, hot or cold compresses, and paracetamol analgesia.
- There is a paucity of evidence on potential side-effects of alternative therapies, e.g. chiropractic and osteopathic manipulation, and further high-quality research is needed to establish whether these therapies are beneficial for low back and/or pelvic pain and safe during pregnancy.
- Exercise in pregnancy has been shown to have other benefits for pregnant women, including reducing excessive gestational weight gain (see Recommendation A.9).

Values

See "Women's values" at the beginning of section 3.D: Background (p. 74).

Resources

Exercise can be administered in a group setting and individually at home; therefore, the cost of exercise interventions varies. Support belts are available commercially from under US$ 10 per item.[5] Physiotherapy and acupuncture require specialist training and are therefore likely to be more resource intensive.

Equity

Improving access to low back and pelvic pain interventions may reduce inequalities by reducing functional disability and sick leave related to low back and pelvic pain among disadvantaged women.

5 Based on Internet search.